Wei-Cheng Lee

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Ph.D. student in Computer Science, RL / online learning theory

Research Interests

I develop learning and optimization algorithms with strong theoretical guarantees for agents that learn while interacting with a changing world, with a focus on online learning, bandits, reinforcement learning, and continual learning. Inspired by AI for math, I also seek a theoretical understanding of how neural networks reason, with the bitter lesson in mind: favoring general methods that scale with computation.

Education

King Abdullah University of Science and Technology (KAUST) Dec. 2025 – Present
Ph.D. Student in Computer Science *Thuwal, Saudi Arabia*

- Advisor: Francesco Orabona.
- Research focus: learning and optimization algorithms with theoretical guarantees; online learning, bandits, reinforcement learning, and continual learning.

King Abdullah University of Science and Technology (KAUST) Aug. 2024 – Dec. 2025
M.Sc. in Computer Science *Thuwal, Saudi Arabia*

- Continued into the KAUST Computer Science Ph.D. program.
- Selected coursework: online learning, high-performance computing, numerical methods for stochastic differential equations, computing systems and concurrency.

National Taiwan University Sep. 2012 – Jan. 2018
B.Sc. in Computer Science, Minor in Mathematics *Taipei, Taiwan*

- Selected coursework: algorithm design and analysis, real analysis, abstract algebra, linear algebra, probability theory, numerical methods.

Publications

A Single Stepsize Suffices for Unprojected Linear TD(0): Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging COLT 2026
Wei-Cheng Lee, Francesco Orabona.

New Lower Bounds for Stochastic Non-Convex Optimization through Divergence Decomposition COLT 2025
El Mehdi Saad, **Wei-Cheng Lee**, Francesco Orabona.

MixLasso: Generalized Mixed Regression via Convex Atomic-Norm Regularization NeurIPS 2018
Ian E.H. Yen, **Wei-Cheng Lee**, Kai Zhong, Sung-En Chang, Pradeep Ravikumar, Shou-De Lin.

Latent Feature Lasso ICML 2017
Ian E.H. Yen, **Wei-Cheng Lee**, Sung-En Chang, Arun S. Suggala, Shou-De Lin, Pradeep Ravikumar.

A Robust $\widetilde{\mathcal{O}}(1/\sqrt{T})$ Rate for Unprojected TD Learning with Linear Function Approximation TMLR
Wei-Cheng Lee, Francesco Orabona.

Best-of-Both-Worlds Causal Bandits with Mediator Feedback EWRL 2026
Hubert Marek Drazkowski, **Wei-Cheng Lee**, Francesco Orabona, Yevgeny Seldin.

A Best-of-Both-Worlds Proof for Tsallis-INF without Fenchel Conjugates arXiv
Wei-Cheng Lee, Francesco Orabona.

Talks

DeLTA Seminar, Department of Computer Science, University of Copenhagen Jun. 2026
A Single Stepsize Suffices for Unprojected Linear TD(0): Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging
Copenhagen, Denmark

Learning Theory Summer School & Workshop Jun. 2026
Contributed talk: A Single Stepsize for Robust and Fast Unprojected TD(0)
Copenhagen, Denmark

Research Experience

Graduate Research, King Abdullah University of Science and Technology (KAUST) Aug. 2024 – Present
M.Sc./Ph.D. Student in Computer Science *Thuwal, Saudi Arabia*

- Proved finite-time high-probability guarantees for unprojected linear TD(0) under Markovian sampling, showing that a single stepsize schedule yields bounded iterates and both robust curvature-free and fast curvature-dependent rates via Polyak–Ruppert averaging.
- Established query-complexity lower bounds for first-order stochastic non-convex optimization under structural conditions such as quasar-convexity, quadratic growth, and restricted secant inequalities, using divergence decomposition to reduce optimization to hard function-identification problems.

- Developed a simplified primal analysis for Tsallis-INF in multi-armed bandits that proves best-of-both-worlds guarantees via OCO tools instead of Fenchel-conjugate machinery.

Institute of Information Science, Academia Sinica Feb. 2022 – Mar. 2024
Research Assistant *Taipei, Taiwan*

- Proved regret guarantees for online reinforcement learning in infinite-horizon average-reward MDPs with full-information feedback, developing a new entropic-linear-program analysis for Hedge-style behavior across stochastic and adversarial regimes.

Graduate Research, National Taiwan University Feb. 2018 – Jun. 2021
M.Sc. Student *Taipei, Taiwan*

- Derived algorithms for log-loss function classes, with emphasis on online portfolio selection.

Engineering Experience

Saudi Aramco, Research & Development Center Jun. 2025 – Jul. 2025
Graduate Intern *Dhahran, Saudi Arabia*

- Prototyped a Pyomo-based solver for gas-oil separation network optimization, formulating the problem as a mixed-integer nonlinear program and integrating JAX-accelerated derivative computations with finite-difference validation.

Graduate Institute of Biomedical Electronics and Bioinformatics, NTU Aug. 2021 – Jan. 2022
Research Assistant *Taipei, Taiwan*

- Contributed to a collaborative NTU-Far Eastern Memorial Hospital project on ICD-10 coding support, applying BERT-based language models to classify clinical case descriptions for Taiwan National Health Insurance workflows.

Teaching

Reinforcement Learning, KAUST Fall 2026, Spring 2026, Spring 2025
Teaching Assistant

Optimization Algorithms, National Taiwan University Fall 2019, Fall 2018
Teaching Assistant

Probability, National Taiwan University Spring 2018, Spring 2015
Teaching Assistant

Honors

Presidential Award, National Taiwan University Spring 2014, Fall 2012
Awarded to the top 5% of students each semester

Technical Stack

Programming: Python, JAX, PyTorch, C/C++, MATLAB, Julia, Go, JavaScript.